

Composition to Verify Inverses - Practice

Algebra II Honors | Unit 1 | Lesson 13 Practice | TEK A2.2.D

Objective: I can compose non-linear pairs and reason about the domain restrictions they require.**Section 1 — Recall**

1. Let $f(x) = 2x + 5$ and $h(x) = 3x$. Compute $f(h(x))$ and $h(f(x))$ as general expressions to see that order matters.

$$f(h(x)) = \underline{\hspace{2cm}} \quad h(f(x)) = \underline{\hspace{2cm}}$$

2. Test $f(x) = 4x - 3$ and $g(x) = (x + 3)/4$ - compute both compositions.

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Test $f(x) = x^2 + 1$ (for $x \geq 0$) and $g(x) = \sqrt{x - 1}$. Compute $f(g(x))$ and $g(f(x))$, being careful about domain.

$$f(g(x)) = \underline{\hspace{2cm}} \quad (x \geq 1) \quad g(f(x)) = \underline{\hspace{2cm}} \quad (x \geq 0)$$

4. Test $f(x) = x^3 - 2$ and $g(x) = \sqrt[3]{x + 2}$. Verify both compositions.

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad (\text{no restriction needed})$$

Section 3 — Challenge

5. For $f(x) = x^2 - 9$ (restricted to $x \geq 0$) and $g(x) = \sqrt{x + 9}$, compute $g(f(-3))$ using the raw rule (ignoring the restriction), and compare the result to -3 itself. What does this show?

$$f(-3) = \underline{\hspace{2cm}} \quad g(0) = \underline{\hspace{2cm}} \quad (\text{not } -3!)$$

6. Prove that if $f(g(x)) = x$ for all x in g 's domain, and $g(f(x)) = x$ for all x in f 's domain, then f and g are truly inverses of each other - explain why both conditions together, not just one, capture the full relationship.

Section 4 — Explain It

7. A classmate claims that because $f(x) = x^2$ and $g(x) = \sqrt{x}$ satisfy $g(f(x)) = x$ only for $x \geq 0$, while $f(g(x)) = x$ works for all $x \geq 0$ in g 's domain, this "asymmetry" means they can never be considered inverses in any meaningful sense. Is the classmate's conclusion too strong? Explain what the restricted relationship actually means.
